Yitian Sun

Trottier Space Institute Postdoctoral Fellow
McGill University, Montréal QC, Canada
yitian.sun.26@gmail.com | yitian.sun@mcgill.ca | github.com/yitiansun

Experience & Education

2024–2026	**Trottier Space Institute Fellow** at McGill University, Canada *Machine Learning and simulation techniques in particle astrophysics and cosmology.*
2024	**Ph.D. in Physics** at Massachusetts Institute of Technology, USA Advisor: Tracy R. Slatyer. GPA: 4.8/5.0. *Thesis: Illuminating the Nature of Dark Matter through Observation, Simulation and Machine Learning.*
2018	**Honors B.A. in Physics** and **B.S. in Mathematics** at University of Chicago, USA GPA: 3.91/4.00.

Programming Languages and Frameworks

Python frameworks: JAX, Tensorflow, PyTorch. Other languages: C/C++, IDL, Mathematica.

Code & Projects

`DM21cm`: main developer. Package for inhomogeneous energy injection in 21-cm cosmology, built with JAX.

`DarkHistory`: contributor. Package for early-universe dark matter energy injection, with neural-network surrogate transfer functions.

`fermi-prob-prog`: contributor. Differentiable probabilistic-programming pipeline for Fermi γ-ray analysis built with Numpyro and JAX.

Reviewer Experience

2023–2025	NeurIPS Workshop on Machine Learning and the Physical Sciences, Reviewer.

Publications

1. S. Mishra-Sharma, T. R. Slatyer, **Y. Sun**, Y. Wu, "High-dimensional inference for the γ-ray sky with differentiable programming," arXiv:2604.08648 (2026).

2. **Y. Sun**, J. W. Foster, J. B. Muñoz, "Constraining inhomogeneous energy injection from annihilating dark matter and primordial black holes with 21-cm cosmology," arXiv:2509.22772 (2025).

3. W. Yang, **Y. Sun**, Y. Wang, *et al.*, "Searching for Axion Dark Matter Gegenschein of the Vela Supernova Remnant with FAST," Astrophys. J. 988, 104 (2025).

4. E. D. Ramirez, **Y. Sun**, M. R. Buckley, S. Mishra-Sharma, T. R. Slatyer, "Inferring the morphology of the Galactic Center excess with Gaussian processes," Phys. Rev. D 111, 063065 (2025).
5. **Y. Sun**, J. W. Foster, H. Liu, J. B. Muñoz, T. R. Slatyer, "Inhomogeneous energy injection in the 21-cm power spectrum: Sensitivity to dark matter decay," Phys. Rev. D 111, 043015 (2025).
6. **Y. Sun**, K. Schutz, H. Sewalls, C. Leung, K. W. Masui, "Looking in the axion mirror: An all-sky analysis of stimulated decay," Phys. Rev. D 109, 043042 (2024).
7. S. Mishra-Sharma, T. R. Slatyer, **Y. Sun**, Y. Wu, "Disentangling γ-ray observations of the Galactic Center using differentiable probabilistic programming," Extended Abstract, ICML 2023 Workshop on Machine Learning for Astrophysics (Spotlight) (2023).
8. **Y. Sun**, T. R. Slatyer, "Modeling early-universe energy injection with dense neural networks," Phys. Rev. D 107, 063541 (2023).
9. **Y. Sun**, K. Schutz, A. Nambrath, C. Leung, K. W. Masui, "Axion dark matter-induced echo of supernova remnants," Phys. Rev. D 105, 063007 (2022).
10. P. Huang, R. A. Roglans, D. D. Spiegel, **Y. Sun**, C. E. M. Wagner, "Constraints on supersymmetric dark matter for heavy scalar superpartners," Phys. Rev. D 95, 095021 (2017).

Presentations

Invited talks

- "Inference, Generative Models, and the Galactic Center Excess", *KITP Workshop on Generative AI for High and Low Energy Physics*, Kavli Institute for Theoretical Physics, UC Santa Barbara, CA (Nov 2025).
- Further invited seminars and colloquia at University of Toronto, Perimeter Institute, Fermilab, University of Florida, Rutgers University, and Oakland University (2023–2026).

Contributed conference & workshop talks

- "Disentangling γ-ray observations of the Galactic Center using Differentiable Probabilistic Programming" (**Spotlight**), *Machine Learning for Astrophysics @ International Conference of Machine Learning (ICML) 2023*, Honolulu, HI (Jul 2023).
- Further contributed talks at international conferences (Pheno, ICHEP, Identification of Dark Matter, TeVPA, PASCOS, Berkeley Axion Workshop, etc.).

Service & Outreach

Conference organization. Organizer, *Dark Side of the Universe* (2025) and *Phase Transitions and Topological Defects in the Early Universe* (2022).
Outreach. Presenter, *Cambridge Science Festival* (2023).